

1 Fig. 1.1 is a diagram drawn from a photomicrograph of a transverse section through part of a leaf.

The arrows in Fig. 1.1 show the movement of water through the cells of the leaf after it has left the xylem.

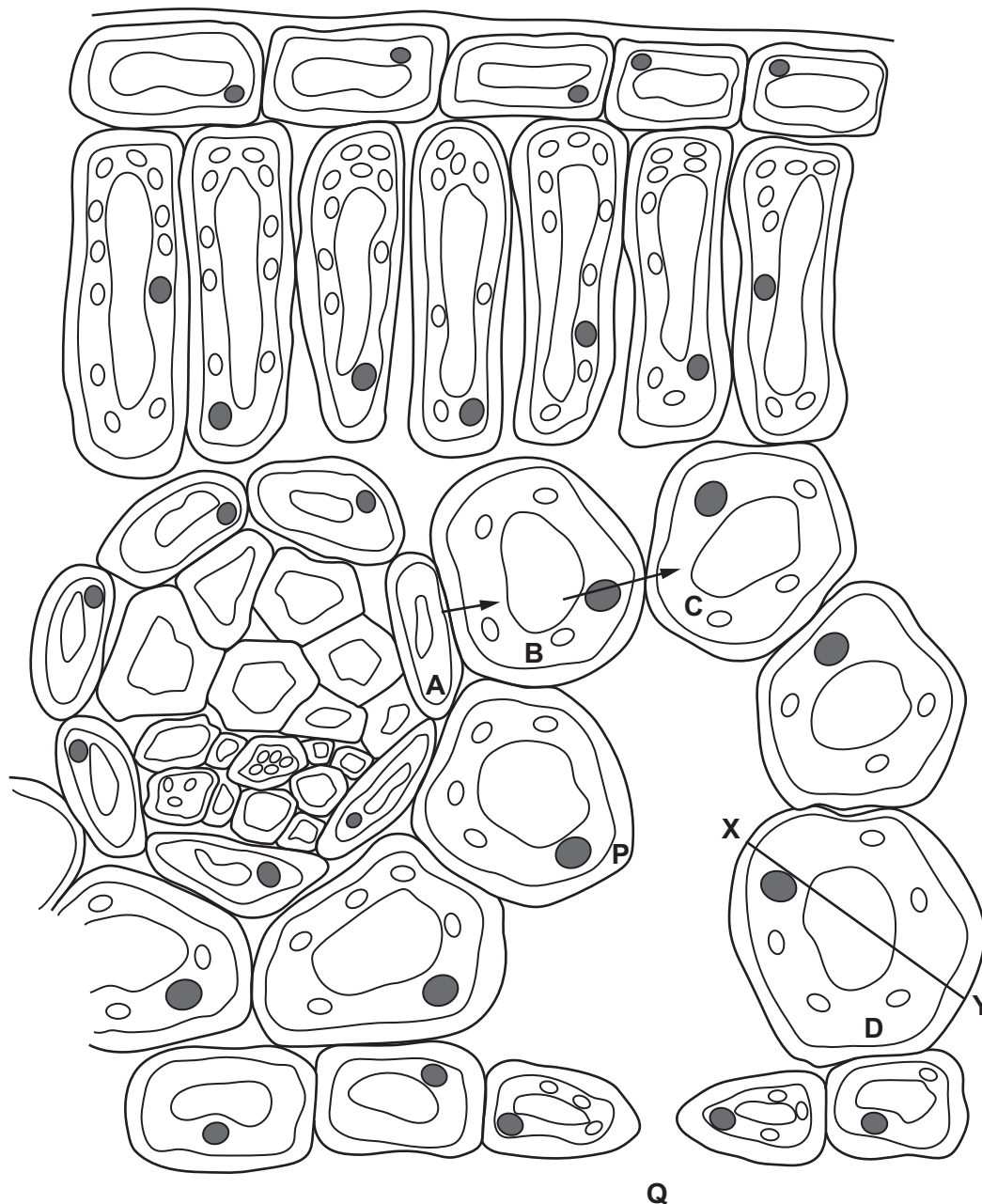


Fig. 1.1

- (a) Water from the xylem can enter cell **A** and then moves to cells **B** and **C** without crossing their cell walls.

The cell structures through which water passes from cell **A** to cell **B** are **not** visible in Fig. 1.1.

- (i) Name the cell structures through which water passes from cell **A** to cell **B** without crossing their cell walls.

..... [1]

(ii) Explain what causes water to move from cell **B** to cell **C**.

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..... [1]

(iii) Name the pathway taken by water between cell **A** and cell **C**.

..... [1]

(b) Most of the water that arrives at the leaf passes to the external atmosphere.

With reference to Fig. 1.1, describe **and** explain the sequence of events occurring between point **P** and point **Q**.

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..... [4]

(c) The actual diameter of cell **D** in Fig. 1.1 along the length **X–Y** is 25 μm .

Calculate the magnification of the image.

Write down the formula used to make your calculation. Show your working.

<i>formula</i>

answer = $\times$ [3]

[Total: 10]